

Single-cell cytokine cascade inference: the coupling fix, why the encoder stays, and what the method can/can't do

In one breath. We infer, from a single snapshot of stimulated immune cells, (i) *which* stimuli are biologically *coupled* (existence) and (ii) *who is upstream* (direction). This note explains the recent coupling-gate fix, shows by ablation that the neural encoder is *not* a removable part, and—the main point—maps where the method works and fails across five datasets, tied to each dataset’s metadata.

1 The coupling fix: what was missing, why it works

The gate decides “are stimuli a, b coupled?”. The old version **over-called** ($\sim 77\%$ of Oesinghaus pairs “coupled”) for two reasons:

- **Over-power.** The significance null was computed at the *cell* level (thousands of cells), so every mean was razor-sharp and almost everything beat the null. The true unit of independence is the *donor* (~ 10), not the cell.
- **Hub/degree bias.** A broadly-engaged signature (e.g. IL-15, induced by many cytokines) looks coupled to *everything*. This lives in the coupling value itself—it survives even a perfect null.

The fix. Work in signature space: $M[a, b] = s(a, S_b) - s(\text{PBS}, S_b)$ (how much a ’s cells express b ’s discovered signature, PBS-normalised). Then **coupling** = $M + M^\top$ (symmetric $\Rightarrow$ existence) and **direction** = $M - M^\top$ (antisymmetric $\Rightarrow$ who is upstream). Replace the cell null with a **donor-level sign-flip null** (unit = donor), and remove hubs by **degree double-centering** $R[i, j] = C[i, j] - d_i - d_j + g$ (d = node strength, g = grand mean). Degree correction is symmetric, so it *only* sharpens existence and never alters direction.

Table 1. Coupling gate on Oesinghaus, fix applied step by step.

Gate variant	null unit	over-call	known-pair recall
old (cell-level null)	cells ($\sim 10^3$)	$\sim 77\%$	8/17
+ donor-level null	donors (~ 10)	53%	8/17
+ degree correction	donors	31%	11/17 (2.1$\times$)

2 Do we need the trained encoder? Two *distinct* questions

There are two separate “do we need the encoder” questions, easily conflated: **(2a)** the signature *source* (does S_X need a trained model, or will model-free DE do?) and **(2b)** the training *procedure* (is the separate cell-type pre-training *stage* needed, or does a single end-to-end train suffice?). They are answered by different experiments.

(2a) Signature source: model attributions vs. differential expression

Holding everything else fixed, we swapped *only* how the signature S_X is defined in a 2×2 {IG, DE} $\times$ {vs-PBS, vs-Panel} design on Oesinghaus (`run_signature_ablation.py`). IG = Integrated Gradients *through the trained model*; DE = differential expression, *no model*.

Verdict (2a): signatures need the trained model. DE drops direction to $\approx$ chance (59% vs 82%) and overlaps IG by only Jaccard = 0.11 (near-disjoint genes), so the model’s attributions carry real signal a DE list lacks. (Corollary: more *specific* signatures (vs-Panel) alone do *not* fix over-call, 76.5 $\rightarrow$ 77.9%—that is the null’s job, §1.) **This compares signature *source* only; it does *not* test whether the cell-type pre-training *stage* is needed—that is (2b).**

Table 2. Signature ablation. DE collapses direction to chance and selects nearly-disjoint genes.

signature	direction acc.	coupled frac.	top hub
IG, vs-PBS (current)	14/17 (82%)	76.5%	IL-15
IG, vs-Panel	14/17 (82%)	77.9%	IL-15
DE, vs-PBS	10/17 (59%)	58.7%	IFN- β
DE, vs-Panel	9/17 (53%)	46.7%	IFN- β

(2b) Training procedure: cell-type pre-training vs. single end-to-end

The pipeline trains in two stages: Stage 1 pre-trains the encoder on cell-type labels, then freezes it and Stage 2 trains the MIL on stimuli. *Is Stage 1 necessary?* We test it head-to-head on Oesinghaus—24 binary models trained two ways with identical data/seed/architecture: **(A)** two-stage (cell-type pre-train $\rightarrow$ frozen MIL) vs. **(B)** single end-to-end (random-init encoder, no cell-type pre-training, encoder unfrozen)—comparing binary validation accuracy and downstream coupling/direction.

Table 2b. Two-stage vs. single-stage (1 seed each; Oesinghaus, 24 cytokines, identical data/seed).

arm	binary val $P(\text{correct})$	direction (/24)	coupling recall (/17)	over-call
two-stage (current)	0.91	15 (62%)	5	0.31
single-stage (no pre-train)	0.88	13 (54%)	11	0.38

Verdict (2b): the pre-training stage is *not* demonstrably load-bearing, but one seed is too noisy to call. Single-stage learns stimulus-vs-PBS essentially as well (val 0.88 vs 0.91) and is competitive downstream—better coupling recall this run, modestly worse direction. Crucially, the freshly-retrained *two-stage* coupling recall (5/17) fell well below its previously-reported 11/17 *at the same over-call*, exposing run-to-run variance (binary training $\rightarrow$ signatures) comparable to the two-stage-vs-single-stage gap itself. So a single end-to-end train is a viable candidate to replace the two-stage pipeline—but a confident “remove Stage-1” decision (and indeed any single-seed coupling number here) needs a multi-seed sweep.

3 The main point: abilities and limits across datasets

All numbers below use the **updated method** (signature coupling + donor/degree fix + cross_asym direction).

Table 3. Dataset metadata (the levers).

dataset	organism / tissue	cells	donors	cell types	labels	panel
Oesinghaus	human PBMC, 24h	$\sim$ 9.7M	12	$\sim$ 18	90 cyt	whole-tx
Sheu	mouse BMDM, 3h	$\sim$ 295K	4*	$\sim$ 3	7 stim	500-gene
Imm. Dict. (Cui)	mouse LN, <i>in vivo</i> 4h	$\sim$ 387K	3/cyt	26	86 cyt	whole-tx
Cano-Gamez	human CD4 T	43K	4	2	4 Th	whole-tx
Oelen	human PBMC	$\sim$ 911K	128	9	6 path $\times$ t	whole-tx

*pseudo-donors. Coupling used a binary-IG subset of labels where training cost required it (Oes 24, Cui 12); whole-tx datasets reduced to 4000 HVG.

[†]latent-geometry coupling *failed* 0/2 on Sheu’s targeted panel; signature-space coupling *rescued* 2/2 IFN cascades. [‡]degree correction has too few nodes (6) to help; symmetric-metric direction control on Cui = 33% confirms the antisymmetric construction is essential.

Table 4. Results with the updated method.

dataset	coupling over-call	coupling recall	direction (cross_asym)	gate path
Oesinghaus	77 $\rightarrow$ 31%	11/17 (2.1 $\times$)	88% (15/17)	donor+degree
Sheu	$\rightarrow$ 43%	4/5 MUST [†]	86% (6/7)	cell+degree
Imm. Dict.	88 $\rightarrow$ 44%	6/7	83% (5/6)	cell+degree
Cano-Gamez	100 $\rightarrow$ 33%	Th17 $\leftrightarrow$ iTreg #1	n/a (parallel fates)	cell+degree
Oelen	33 $\rightarrow$ 47% [‡]	1/3	progression 2/3	donor @128

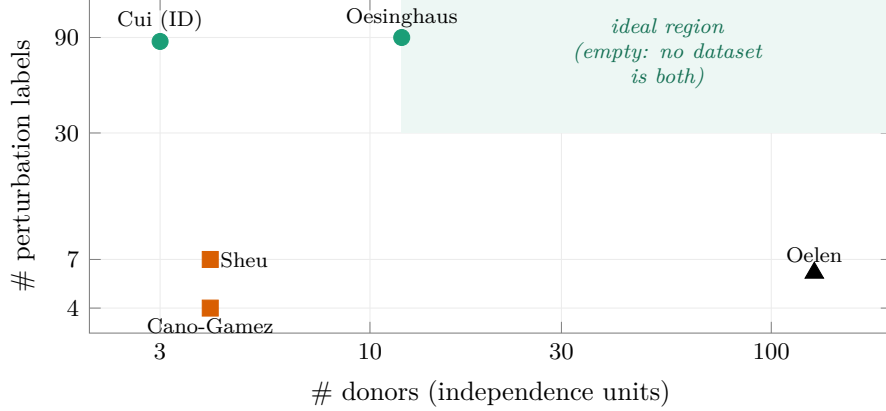


Figure 1. The method needs *both* many donors (powered null) *and* many labels (stable degree correction). **Green:** coupling+direction work; **orange:** partial; **blue:** donor-scale milestone but too few labels. The joint “ideal” corner is empty—the standing gap.

3.1 One line per dataset

- **Oesinghaus** — 90 cytokines on human PBMC; the method’s home turf (enough labels *and* donors). Best result: over-call halved to 31% with rising recall; direction 88%.
- **Sheu** — 7 TLR/cytokine stimuli on a *500-gene targeted* mouse-BMDM panel. The narrow panel makes everything co-vary, so latent-geometry coupling fails (0/2); signature-space coupling rescues the textbook IFN cascades (2/2); direction 86%.
- **Cui / Immune Dictionary** — 86 cytokines, mouse lymph node *in vivo*. Whole-transcriptome with many labels $\Rightarrow$ clean coupling (88 $\rightarrow$ 44%, recall 6/7) and direction (83%), even at only 3 donors (cell-level path).
- **Cano-Gamez** — 4 T-helper polarisation conditions on human CD4 T cells. Recovers the known Th17 $\leftrightarrow$ iTreg (TGF- β) coupling as #1, but the 4 “labels” are parallel fates, not a cascade, so direction is undefined.
- **Oelen** — 3 pathogens $\times$ 2 time-points on human PBMC with **128 donors**. First time the donor-level null ran fully powered ($\sim$ 100 donors/pair). But 6 labels are too few for degree correction, and pathogens share one innate program, so the coupling benchmark is weak (1/3); the 3h $\rightarrow$ 24h progression direction is 2/3.

3.2 Why they differ — four metadata-driven rules

1. **# labels governs degree correction.** Works at $\gtrsim$ 15–20 conditions (Oes, Cui); *fails* at 4–6 (Cano, Oelen)—too few nodes to estimate hub strength (over-call even rose on Oelen).
2. **# donors governs the gate path.** <8 donors $\Rightarrow$ cell-level fallback (Sheu, Cui, Cano); $\geq 8 \Rightarrow$ powered donor-level null; Oelen’s 128 is the scale proof.
3. **Panel breadth governs the shared-activation confound.** A targeted 500-gene panel (Sheu)

collapses latent geometry; signature space (specific genes) is panel-robust.

4. **Label semantics govern whether *direction* exists.** Cytokine perturbations with known cascades $\Rightarrow$ direction (83–88%); parallel fates (Cano) or pathogens (Oelen) give coupling/progression only, never cytokine $\rightarrow$ cytokine direction.

Net. The method needs *many labels* (degree correction) + ≥ 8 *donors* (powered null) + *perturbation labels with directional relationships*. Oesinghaus uniquely has all three; every other dataset is missing one—exactly where it underperforms. The open need is a dataset that is simultaneously high-donor *and* many-condition *and* discriminative.

4 Datasets without updated-method results (before this update)

COVID-19 (Stephenson/Haniffa) — not a valid result yet. We ran `cross_asym` to recover disease-*progression* direction (less-severe = upstream) from ~ 780 K PBMCs over 5 severity grades (130 patients). The method’s apparatus *passed* its go/no-go gate (distinct-program ladder 100%; monotone-intensity ladder 0%, flagging the magnitude-confound regime). **But** the grade-vs-Healthy IG signatures were *confounded* (cell-cycle genes, CITE-seq antibody tags, hemoglobin), so `cross_asym` carried no severity signal (cross-accuracy $0.44 <$ symmetric control 0.67 ; Kendall $\tau = -0.2$). The **coupling degree-fix and donor-level null were never applied here**. Treat as “before this update”; a valid attempt requires cleaned signatures (drop ADT/hemoglobin/cell-cycle, grade-vs-grade contrasts) and the donor-level null.

Datasets & references

- Oesinghaus et al. (2024/25), *A single-cell cytokine dictionary of human peripheral blood* — 9.7M PBMC, 90 cytokines. [Parse 10M PBMC data](#); [bioRxiv 2025.12.12.693897](#).
- Sheu et al. (2024), *Molecular Cell* — mouse BMDM, 500-gene panel. [GEO GSE224518](#).
- Cui et al. (2024), *Nature* 625:377 — Immune Dictionary, 86 cytokines (mouse). [doi:10.1038/s41586-023-06816-9](#); [SCP2554](#).
- Cano-Gamez et al. (2020), *Nature Communications* 11:1801 — CD4 T-cell cytokine responses. [doi:10.1038/s41467-020-15543-y](#); [BioStudies S-BSST2978](#).
- Oelen et al. (2022), *Nature Communications* 13:3267 — 1M-scBloodNL, 120-donor PBMC pathogen stimulation. [doi:10.1038/s41467-022-30893-5](#); [eQTLGen 1M-scBloodNL](#).
- Stephenson et al. (2021), *Nature Medicine* 27:904 — COVID-19 PBMC atlas, 130 patients. [doi:10.1038/s41591-021-01329-2](#).